

Prove each of the statements in 10-18 by mathematical induction.

5.2.14.
$$\sum_{i=1}^{n+1} i * 2^i = n * 2^{n+2} + 2, \text{ for every integer } n \geq 0.$$

$$P(n) : \sum_{i=1}^{n+1} i * 2^i = n * 2^{n+2} + 2$$

Proof:

Basic step $P(0)$:

$$\sum_{i=1}^{(0)+1} i * 2^i = \sum_{i=1}^1 i * 2^i$$

Which means that the lower bound is 1,
and the upper bound is 1. So from $i = 1$
to $i = 1$.

For the left side:

$$\sum_{i=1}^1 i * 2^i = (1) * 2^{(1)} = 1 * 2 = 2$$

And for the right side:

$$n * 2^{n+2} + 2 = (0) * 2^{(0+2)} + 2 = 0 * 4 + 2 = 2$$

Therefore, $P(0)$ is true, for every integer $n \geq 0$.

Inductive step $P(k+1)$:

Suppose $P(k)$ is true.

$$P(k) : \sum_{i=1}^{k+1} i * 2^i = k * 2^{k+2} + 2$$

We must show that $P(k+1)$ is true.

$$P(k+1) : \sum_{i=1}^{k+2} i * 2^i = (k+1) * 2^{k+3} + 2$$

Since $P(k)$ can be written as:

$$\sum_{i=1}^{k+1} i * 2^i = 1 * 2^2 + 2 * 2^3 \dots + (k+1) * 2^{k+1}$$

We can write $P(k+1)$ as:

$$\sum_{i=1}^{k+2} i * 2^i = 1 * 2^2 + 2 * 2^3 \dots + (k+1) * 2^{k+1} + (k+2) * 2^{k+2}$$

$$\sum_{i=1}^{k+2} i * 2^i = \sum_{i=1}^{k+1} i * 2^i + (k+2) * 2^{k+2}$$

$$k * 2^{k+2} + 2 + (k+2) * 2^{k+2} = 2^{k+2}(k+k+2) + 2$$

$$2^{k+2}(k+k+2) + 2 = 2^{k+2}(2k+2) + 2$$

$$2k+2 = 2(k+1)$$

$$2^{k+2}(2k+2) + 2 = 2^{k+2}(2(k+1)) + 2$$

$$2^{k+2} * 2^1 = 2^{k+3}$$

$$2^{k+3}(k+1) + 2 = (k+1) * 2^{k+3} + 2$$

Therefore, the left side is proven for $P(k+1)$.

And for the right side:

$$n * 2^{n+2} + 2 = (k+1) * 2^{((k+1)+2)} + 2 = (k+1) * 2^{k+3} + 2$$

$$(k+1) * 2^{k+3} + 2 = 2^{k+3}(k+1) + 2$$

Which is what was to be shown.

□

Solution: $\sum_{i=1}^{n+1} i * 2^i = n * 2^{n+2} + 2$, for every integer $n \geq 0$ is true.